

INTRODUCTION TO R FOR RESEARCHERS



CSTAT

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AGENDA

1. What is R?

2. Why R?

3. How to use R?

I. Installation

II. Some Concepts and good practices

4. Hands on excises.

I. Data Preparation

II. Initial Data Analysis (Descriptive statistics)



WHAT IS R?

Statistical Programming Language

R is an interpreted programming language for statistical computing and graphics. Created by statisticians **R**oss Ihaka and **R**obert Gentleman and first shared in 1993. It considered to be successor of S language created by AT&T Bell laboratories.

The Comprehensive R Archive Network (**CRAN**) was founded in 1997 by Kurt Hornik and Fritz Leisch to host R's source code, executable files, documentation, and user-created packages.



WHY R?

Great research tool

- 1.**Free!** Every one can use it and every one can contribute.
- 2.**Advanced statistical methods.** Not available in other softwares.
- 3.**Flexible.** Simple calculations or complex data analysis report.
- 4.**Powerful graphics.** No need to manually edit figures.
- 5.**Render multiple output formats.** Can be interactive format.
- 6.**Reproducible Research.** Benefit the entire scientific research community.

... And many more.



HOW TO USE R?

Install R

Installing R

It is straightforward installing R on your machine. Follow these steps:

1. Go to the CRAN (Comprehensive R Archive Network) [R website](#).
2. Choose to download R for Linux, Mac or Windows. Latest version: Latest version: R-4.6.0 (released on 2026-04-24, Because it was There).
3. For Windows users, just install 'base' and this will link you to the download file
4. For Mac users, make sure you download the right version for your CPU.



HOW TO USE R?

Install IDE: RStudio

Installing RStudio

Rstudio is the most popular integrated development environment (IDE) for R. It makes working with R easier, more productive, and organized, especially for new users.

Install steps:

1. Go to the RStudio download website (Posit).
2. Choose the 'Installers for your system' to download
3. Install RStudio as per the instructions.



HOW TO USE R?

Cloud options: Posit
cloud

Posit cloud offers cloud solution to use R, Rstudio without installing on local machine.

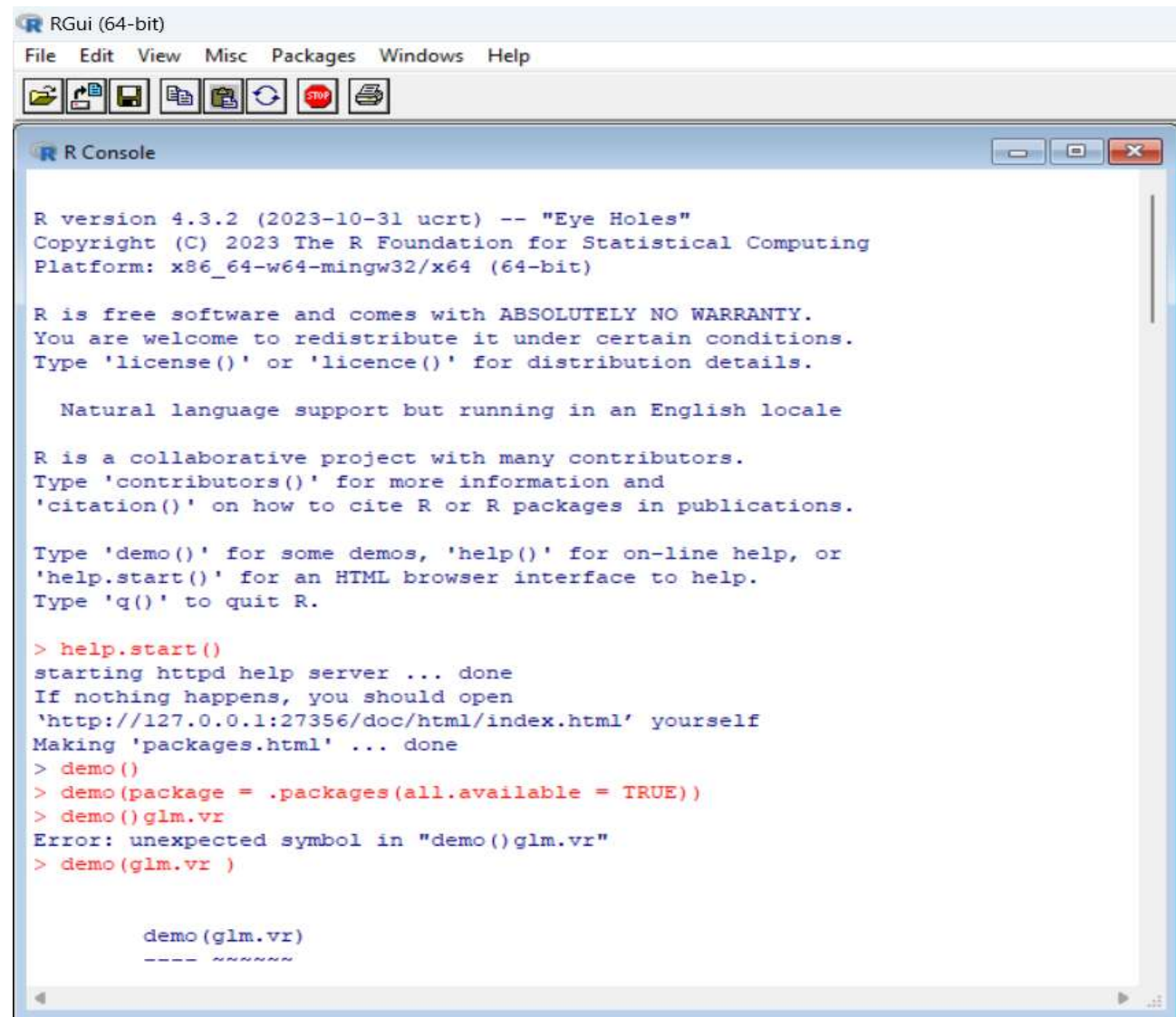
Follow steps:

1. Go to website Posit cloud to register new user/login. Different plans with different limits. Free one limit 25 projects.
2. Create new workspace.
3. Click create new project.



HOW TO USE R?

R GUI



The screenshot shows the RGui (64-bit) window with the R Console pane active. The console displays the R version information, copyright notice, and a list of commands and their outputs. The commands entered are `help.start()`, `demo()`, `demo(package = .packages(all.available = TRUE))`, `demo() glm.vr`, and `demo(glm.vr)`. The output for `demo()` shows the starting of the httpd help server. The output for `demo(glm.vr)` shows the start of the glm.vr demo.

```
RGui (64-bit)
File Edit View Misc Packages Windows Help

R Console

R version 4.3.2 (2023-10-31 ucrt) -- "Eye Holes"
Copyright (C) 2023 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> help.start()
starting httpd help server ... done
If nothing happens, you should open
'http://127.0.0.1:27356/doc/html/index.html' yourself
Making 'packages.html' ... done
> demo()
> demo(package = .packages(all.available = TRUE))
> demo() glm.vr
Error: unexpected symbol in "demo() glm.vr"
> demo(glm.vr )

demo(glm.vr)
---- ~~~~~
```



HOW TO USE R?

R studio Layout

Tools>global options to change layout and settings

The screenshot displays the RStudio interface with three main panes highlighted by colored boxes and red text labels:

- Newly created R markdown file**: The Source pane (top left) shows a new R Markdown document titled "Intro to R". It contains a title, output format (html_document), date (2024-01-16), and several code chunks. The first chunk is an R setup chunk, and the second is an R Markdown chunk. The third chunk is an R code chunk that generates a summary of the 'cars' dataset. The fourth chunk is an R code chunk that generates a plot of 'pressure'.
- Console**: The Console pane (top right) shows the output of the R code executed in the Source pane. It displays the R version (4.3.1), copyright information, and the output of the 'summary(cars)' and 'plot(pressure)' functions.
- Files/Environment**: The Environment pane (bottom right) shows the current environment, which is empty. The Files pane (bottom left) shows the file structure of the project, including the 'Intro to R' file.

The History pane (bottom left) shows the sequence of R commands executed in the console, including the generation of random numbers, data frame creation, and plotting functions.



HOW TO USE R?

Some concepts

- **Packages.** a collection of functions, documentation, and data.
`install.packages(pkgnames), library(pkgnames)`
- **Functions.** Get things done (outputs). Eg, `mean()`, `sd()`, inputs of functions called arguments
- **Object-oriented programming.** The output can be used as input,
Assignment: `<-` (alt+-)
- **Case sensitivity.** It treats great and Great differently. `class()` or `typeof()`
- **Data types.** numeric(real or decimal), integer, character, complex, and logical.
- **Data structures.** Vector, matrix, factor, list, data frame. `class()`



HOW TO USE R?

Good practices

- **Start with a clean environment.** Recommend to create a project, do not use `setwd()`.
- **Consistent coding style.** Google's R Style Guide. Eg: name convention(not start with number), indentation, etc.
- **Comment the code.** Literate programming. code is explained. Comment starts with `#`
- **Use pipe** `|>`.
- **Use shortcut keys.** Tools → Keyboard Shortcuts Help or Alt + Shift + K.
`alt+enter`: run code. `Ctrl+alt+i:insert`: chunk. `Ctrl+shift+M`: `|>`.
- **help(functionname) or ?functionname.** Get information of any specific named function
- **Github.** Version control is a time machine.
- **Rmarkdown or Quarto.** Updates!!!! and many more...



RESOURCES

BOOKS

- R for Data Science by Hadley Wickham
 - ggplot2: Elegant Graphics for Data Analysis by Hadley Wickham
 - Advanced R by Hadley Wickham
 - R Graphics Cookbook by Paul Teetor
 - R in a Nutshell: A Desktop Quick Reference by Joseph Adler
 - R in Action by Robert Kabacoff
 - Introductory Statistics with R by Peter Dalgaard
 - R for Everyone: Advanced Analytics and Graphics by Jared P. Lander
- ... and many more.



RESOURCES

WEBSITES

- An Introduction to R (<https://cran.r-project.org/doc/manuals/R-intro.pdf>)
- R-bloggers (<https://www.r-bloggers.com/>)
- R pub (<https://rpubs.com/>)
- R-seek (<https://rseek.org/>)
- Data carpentry lessons (<https://software-carpentry.org/lessons/>)
- Quick-R (<https://www.statmethods.net/>)
- MSU R Programming for Data Sciences course online book (jeffdoser.com)
- The R graph gallery (<https://r-graph-gallery.com/>)
- Good practices (<https://swcarpentry.github.io/good-enough-practices-in-scientific-computing/>)





HANDS-ON

DEMOS

DATA PREPARATION



This class assume we all have clean data, which means the data is ready for analysis.

- Do not underestimate the time for data preparation. Often than not most of the time will be spent here.
- Tidy data concept. Please refer to Wickham 2014. 3 Features of **tidy data**:
 - 1.Each variable form a column.
 - 2.Each observation forms a row
 - 3.Each type of observational unit forms a table.
- package tidyverse is a great help.

References: Wickham, Hadley. 2014. "Tidy Data". *Journal of Statistical Software* 59(10):1-23.
<https://doi.org/10.18637/jss.v059.i10>.

DATA PREPARATION

□ Messy Dataset Features:

1. Headers are values ('Day 1', 'Day 2')
2. Rows are variables ('Tmax', 'Tmin')
3. Cells contain merged data ('Site_Country')

Site_Country	Element	Day 1	Day 2
10A_US	Tmax	22°C	24°C
10A_US	Tmin	10°C	11°C
21B_CA	Tmax	15°C	14°C

1. Pivot Longer
(Days to Rows)

2. Separate Column
(Split ID & Country)

3. Pivot Wider
(Elements to Headers)

Tidy Dataset Rules Met:

Each unique variable isolated into its own explicit column header.

Site	Country	Day	Tmax	Tmin
10A	US	Day 1	22°C	10°C
10A	US	Day 2	24°C	11°C
21B	CA	Day 1	15°C	NaN
21B	CA	Day 2	14°C	NaN

INITIAL DATA ANALYSIS (IDA)



IDA primarily ensures transparency and integrity of preconditions to conduct appropriate statistical analyses in a responsible manner to answer predefined research questions [1]. It includes metadata setup, **data cleaning**, data screening, Documentation and initial data reporting, Refining and updating the statistical analysis plan [2].

Data screening is related to descriptive statistics. It includes:

1. Distribution of single variables.
2. Missing data.
3. Association between variables.

References:

- [1]. Baillie M, le Cessie S, Schmidt CO, Lusa L, Huebner M, Topic Group "Initial Data Analysis" of the STRATOS Initiative. Ten simple rules for initial data analysis. PLoS Computational Biology. 2022;18(2):e1009819
- [2]. Huebner, M., le Cessie, S., Schmidt, C.O., & Vach, W. (2021). A Contemporary Conceptual Framework for Initial Data Analysis. *Observational Studies*, 4, 171 - 192.

INITIAL DATA ANALYSIS (IDA)



```
remotes::install_github("higgi13425/medicaldata")
```

Acute Bacterial Meningitis Dataset:

I will use ABM dataset to demonstrate descriptive data analysis. Please use own clean dataset to follow along.

We will move to rstudio workspace.

References: Spanos A, Harrell FE Jr, Durack DT. Differential diagnosis of acute meningitis. An analysis of the predictive value of initial observations. JAMA. 1989 Nov 17;262(19):2700-7. PMID: 2810603.